An Energy-Based JEPA World Model of Drug Perturbations

Predicting single-cell state in latent space on Tahoe-100M and MicrobeAtlas

Dyn**AMI**cs

TL;DR : A true Energy-Based JEPA for drug response.

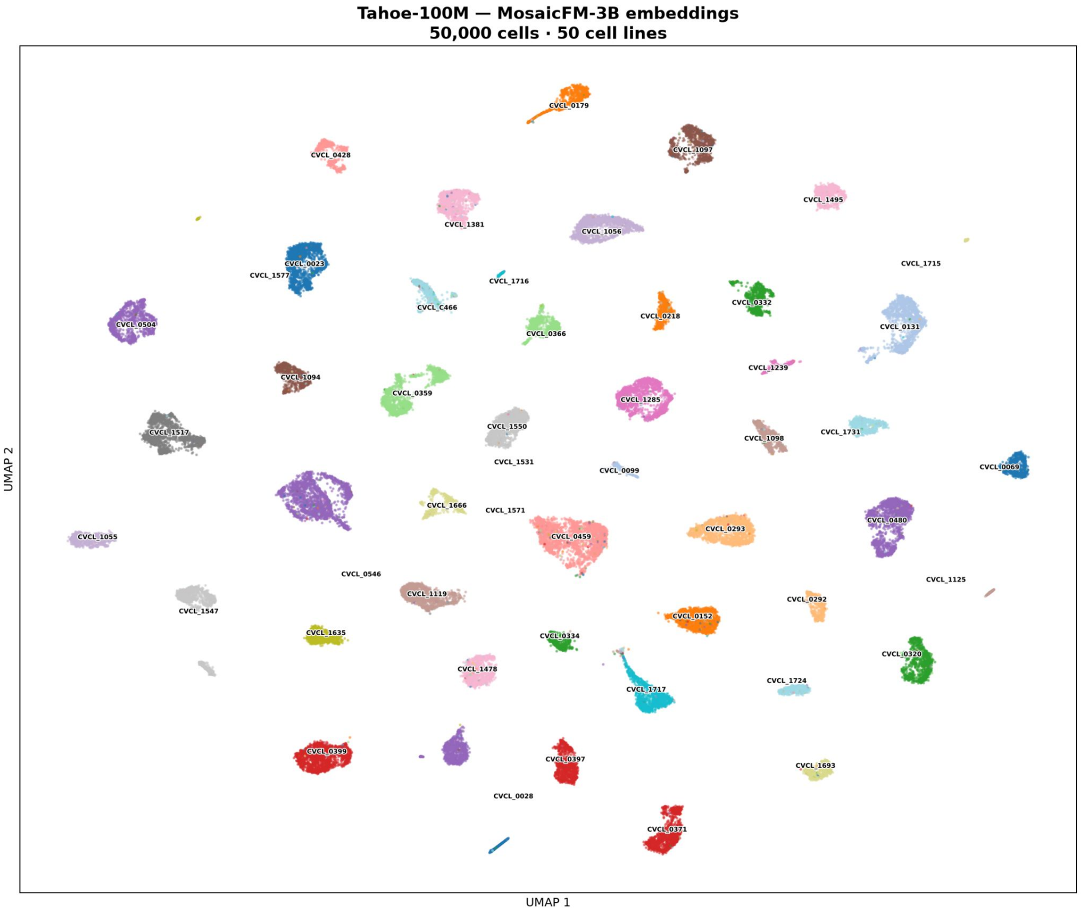
From biological snapshots to controllable world models, Learns the latent dynamics of living systems under intervention.

- Single-cell transcriptomes are noisy, sparse, high-dimensional sets of genes. Reconstructing counts spends capacity on task-irrelevant noise, wasteful.

- Today's transcriptomic foundation models **reconstruct counts** (MLM/next-token) → they spend capacity modeling **task-irrelevant noise**.

- The drug signal is weak and buried under cell identity

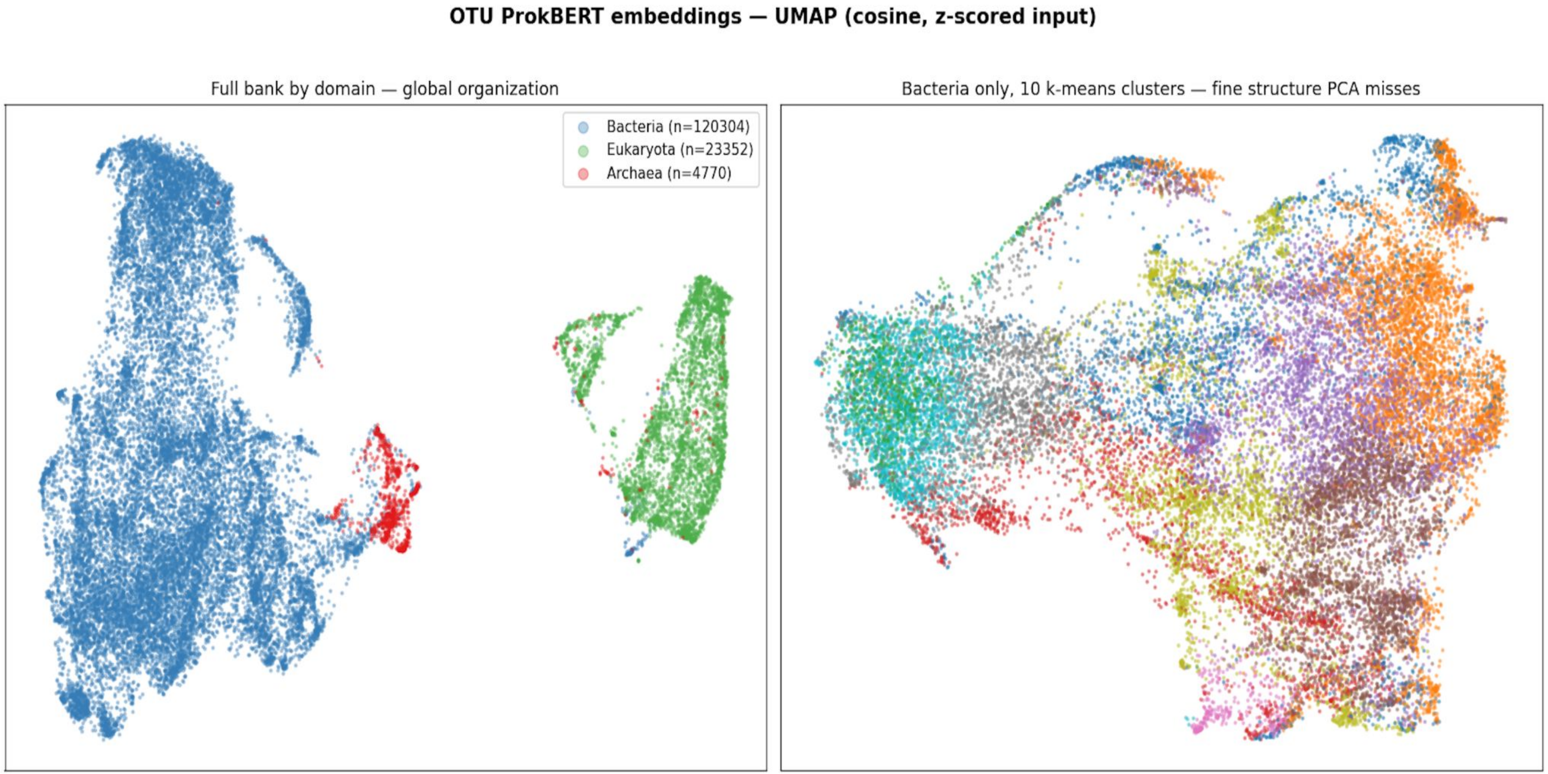
Tahoe 100-M



100M+ single-cell transcriptomic profiles

50 cancer cell lines

MicrobeAtlas

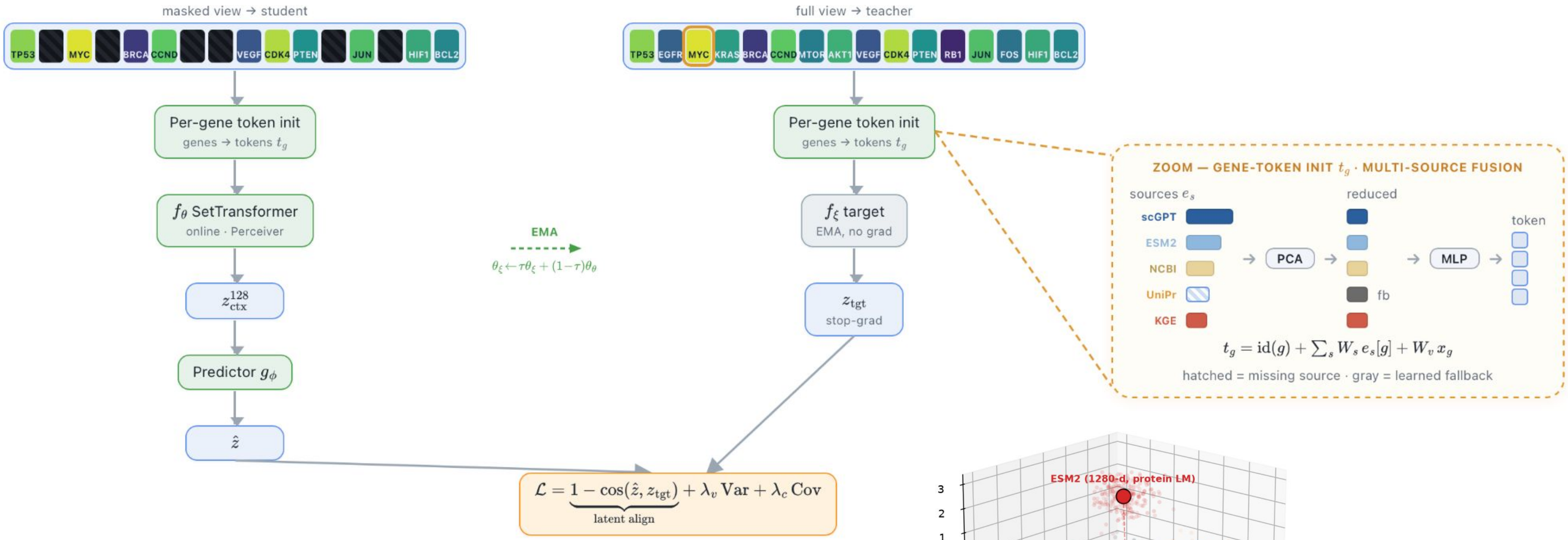


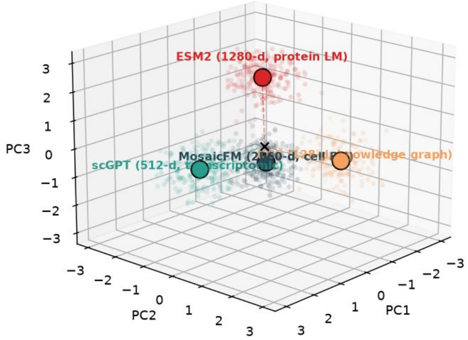
150K+ taxa

2.4M microbiome samples

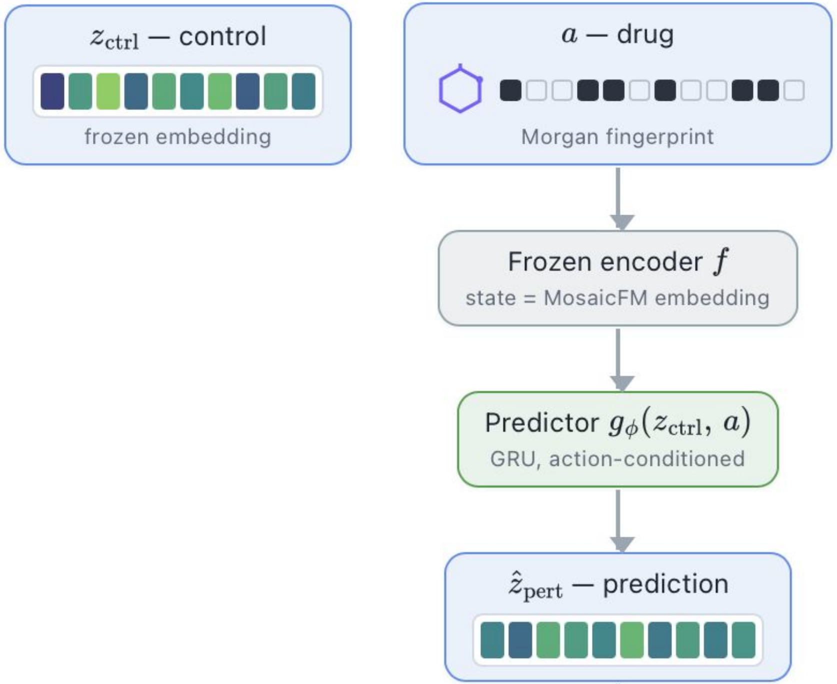
6.9T sequencing reads harmonized

Step 1: Grounded masked-gene

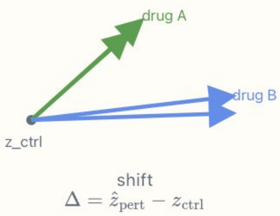




Step 2: Drug Perturbation World Model



③ Perturbation-signature loss — drug-specific direction



$\Rightarrow$ $\mathcal{L}_{\text{sig}} = \text{mean}_{d_i = d_j}\left(1 - \cos(\Delta_i, \Delta_j)\right)$

same drug → same direction (cos→1)

Captures the **drug signature**, not cell-specific noise.

$$\mathcal{E} = \|g_\phi(z_{\text{ctrl}}, a) - z_{\text{pert}}\|^2 \;+\; \lambda_{\cos}\big(1 - \cos(\hat{z}_{\text{pert}}, z_{\text{pert}})\big) \;+\; \lambda_s\,\mathcal{L}_{\text{sig}} \;+\; \lambda_p\,\mathcal{L}_{\text{path}}$$

Keeping the action : slow vs fast features

THE PROBLEM

JEPA latent prediction drifts toward **slow features** (Sobal et al., 2022):

slow — the community's static identity / initial basin $\rightarrow$ **kept**

fast — the state change and the applied **action** a_t $\rightarrow$ **dropped first**

SOLUTION — INVERSE-DYNAMICS MODEL

Recover the action from two consecutive latents $z_t = f_\theta(s_t)$:

$$\hat{a}_t = h_\psi(z_t,\, z_{t+1}) \qquad\qquad \mathcal{L}_{\text{IDM}} = \|\hat{a}_t - a_t\|^2$$

If the encoder dropped a_t, the IDM cannot recover it $\Rightarrow$ the gradient forces f_θ to keep it.

MEASURE — COLLAPSE PROBE

Fresh ridge regression on the **frozen** encoder:

$$\hat{a}_t = W[\, z_t;\, z_{t+1} \,] \qquad\qquad R^2_{\text{action}} \qquad\qquad f_\theta^{\text{rand}} \Rightarrow R^2 \approx 0$$

Evaluating the representation

ACTION RECOVERY

Decode the action a_t from $[\, z_t;\ z_{t+1}\,]$ — frozen encoder, linear probe.

R^2: 0.52 $\rightarrow$ **0.748** with IDM (collapse regime)

DOWNSTREAM — INFANT ENVIRONMENT, 12 CLASSES (AUC)

frozen JEPA + linear probe	**0.896**
supervised MLP (Susagi)	**0.890**
+ SIGReg probe / fine-tuned	**0.894 / 0.918**

→ **frozen self-supervised ties supervised; fine-tuning beats it.**

On Tahoe — frozen encoder already strong → IDM **hurts** (R^2 0.20 → 0.02); helps only on collapse.

Sequencing — protocol stays linearly separable (bal-acc 0.952): the latent keeps the nuisance signal.

Intervention Planning

From a start s_0, reach a target community $s^\star$ by optimizing actions $a_{0:T-1}$ — minimize a latent energy to the target (MPPI):

$$\min_{a_{0:T-1}} \|z_T - z^\star\|^2 \quad \text{s.t.} \quad z_{t+1} = g_\phi(z_t, a_t), \quad z^\star = f_\theta(s^\star)$$

energy = squared latent distance to the target

Pure-JEPA latent-MPPI: 0% success

1 · controllability ✗→✓

need full actuation

oracle fails at K=6 (final 4.09); succeeds at K=24 (0.79).

2 · dynamics ✓

the model is faithful

latent rollout diverges only ~2% over 20 steps.

3 · geometry ✗

distance is uninformative

corr(latent dist, true dist) ≈ 0 — the wall.

Closing the loop

The wall is the **geometry**: latent distance ≠ the distance we plan with.

IDEA — MAKE THE LATENT ISOMETRIC

Add a loss so latent distances reflect true-state distances (up to a scale α):

$$\mathcal{L}_{\text{iso}} = \left(\|z_a - z_b\| - \alpha \, \|s_a - s_b\| \right)^2$$

WE TRIED

✗ stronger IDM (self-supervised) → 0%

✗ smaller latent dimension → 0%

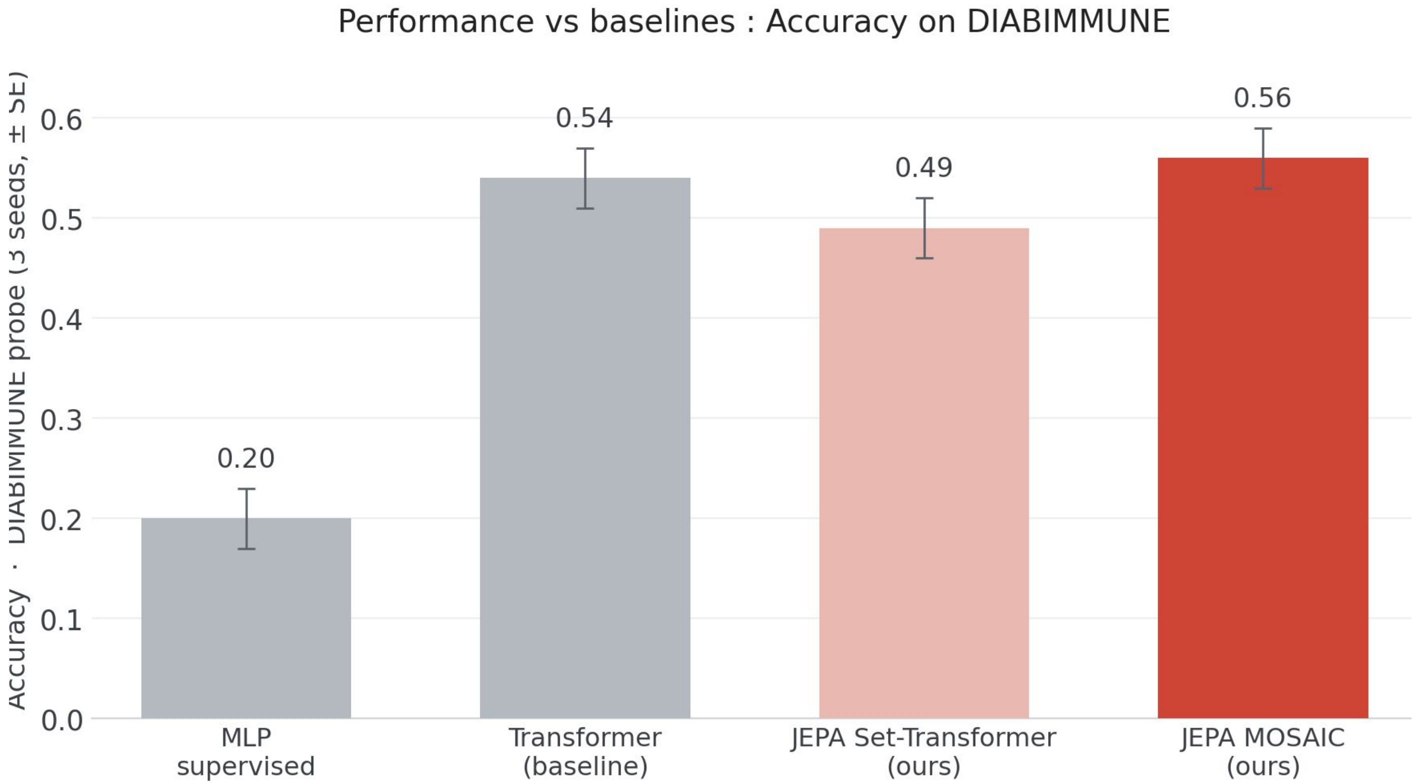
✗ SIGReg isotropy — fixes variance, not the metric → 0%

✓ the isometry loss — the latent becomes a metric → **planning works**

final distance 0.80 vs oracle 0.79

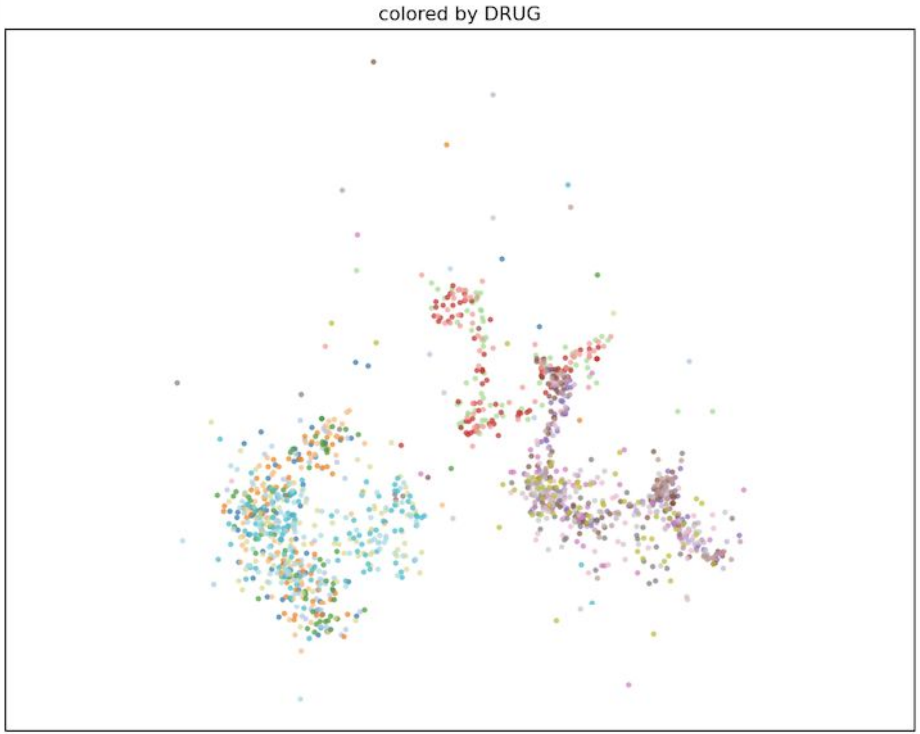
tolerance 1.0 (start ≈ 6.6) — 100% across seeds and 4 different gLV systems.

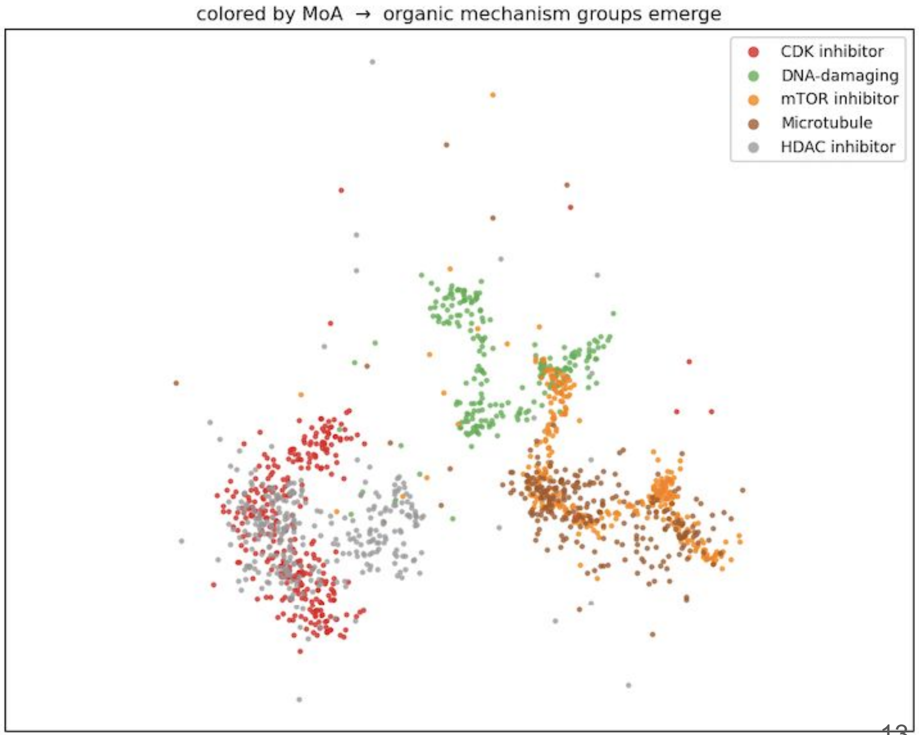
Evaluation



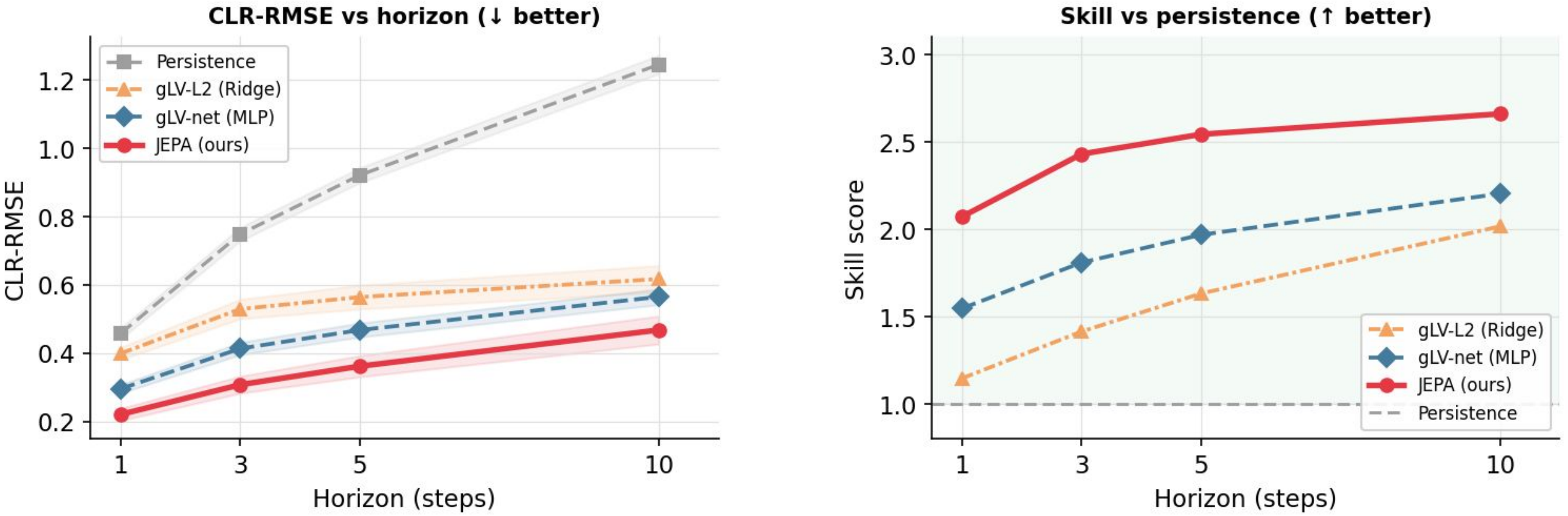
Performance vs baselines : Accuracy on DIABIMMUNE
Accuracy · DIABIMMUNE probe (3 seeds, ± SE)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.20
0.54
0.49
0.56
MLP supervised
Transformer (baseline)
JEPA Set-Transformer (ours)
JEPA MOSAIC (ours)

Predicted perturbations organize by mechanism of action, not by individual drug



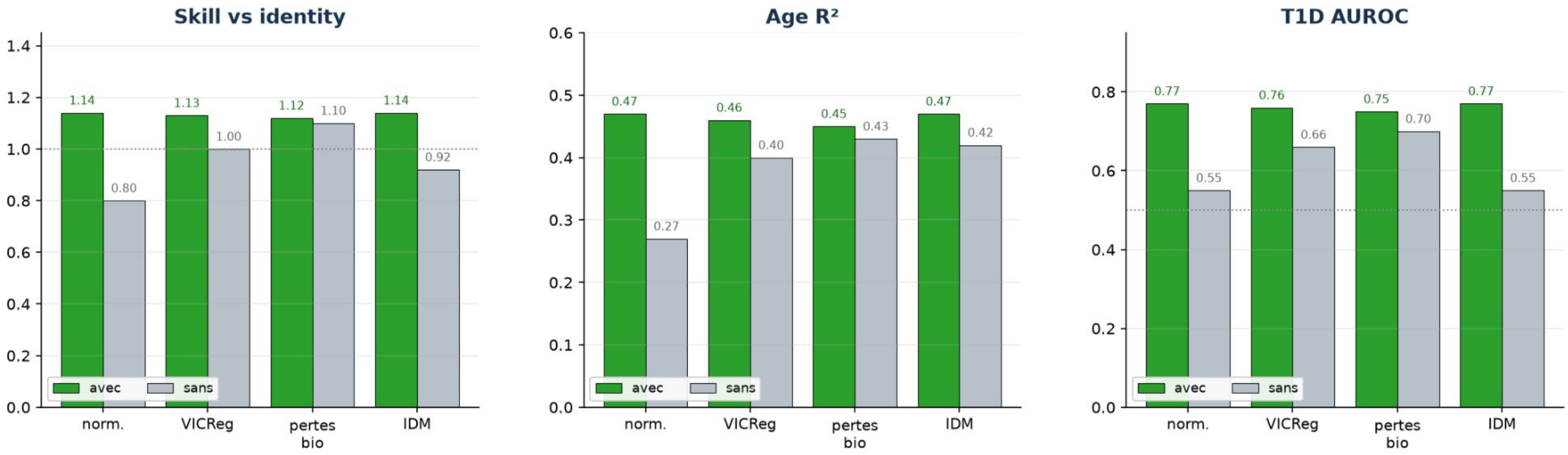


Evaluation

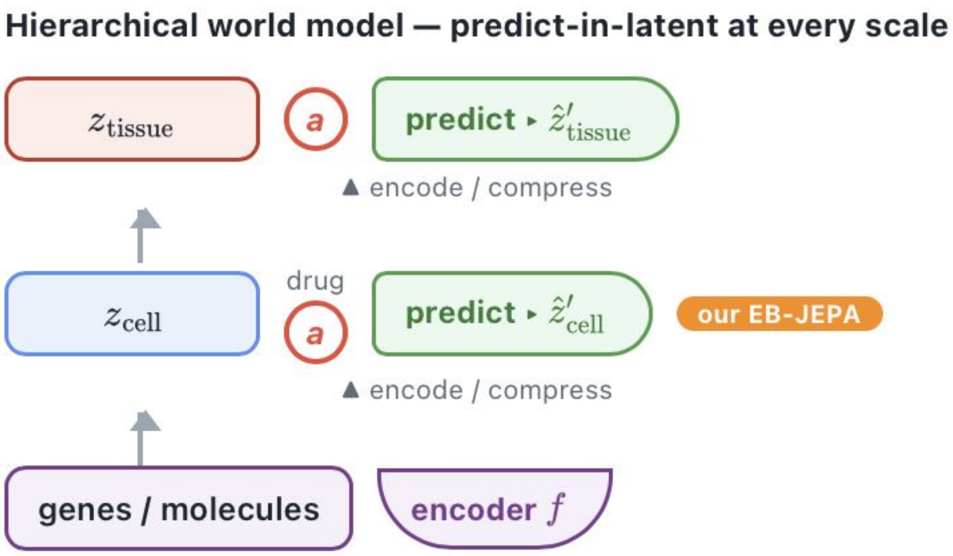


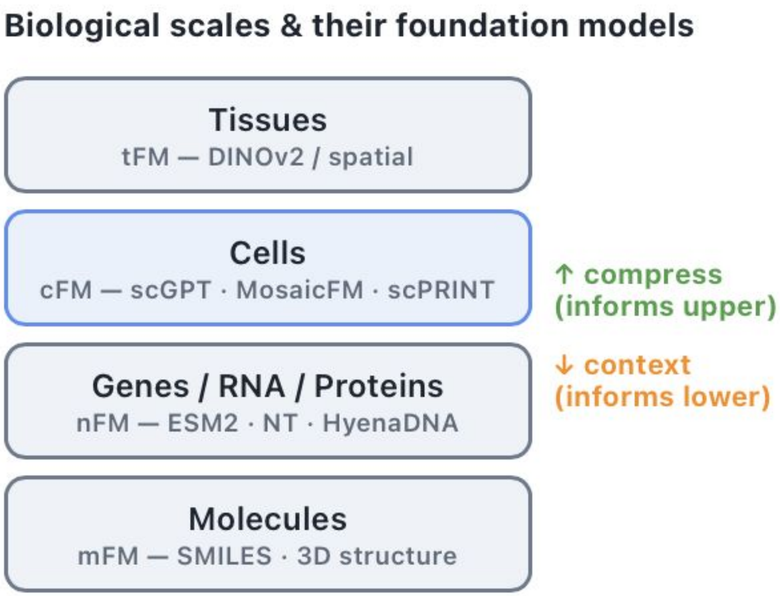
CLR-RMSE vs horizon (↓ better)
Persistence
gLV-L2 (Ridge)
gLV-net (MLP)
JEPA (ours)
CLR-RMSE
0.2
0.4
0.6
0.8
1.0
1.2
1
3
5
10
Horizon (steps)
Skill vs persistence (↑ better)
Skill score
1.0
1.5
2.0
2.5
3.0
gLV-L2 (Ridge)
gLV-net (MLP)
JEPA (ours)
Persistence
1
3
5
10
Horizon (steps)

Ablations


Skill vs identity
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.14
0.80
1.13
1.00
1.12
1.10
1.14
0.92
avec
sans
norm.
VICReg
pertes bio
IDM
Age R²
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.47
0.27
0.46
0.40
0.45
0.43
0.47
0.42
avec
sans
norm.
VICReg
pertes bio
IDM
T1D AUROC
0.8
0.6
0.4
0.2
0.0
0.77
0.55
0.76
0.66
0.75
0.70
0.77
0.55
avec
sans
norm.
VICReg
pertes bio
IDM

A Hierarchical, Cross-scale world model of Biology





Thanks